DAY 22 | HANDS-ON ACTIVITIES

Embeddings, Vector DBs and the RAG App: Problem Statements

Seven problems on the handbook plus IT and travel FAQs. Offline embeddings are lexical stand-ins; add a key for real semantic embeddings. Offline scores show how the code works, not real model quality.

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How to work: read the problem, write the steps in plain English first, then write the code. Test with the sample run, then try your own inputs, including wrong ones.

Problem 1: Similarity Matrix

Level: Warm-up | **Concepts:** embeddings, cosine similarity and distance

Embed 6 sentences (two about leave, two about travel, two about passwords) and print the cosine similarity and distance matrices. Which pairs are most similar?

Sample run / expected result

```
leave-1 vs leave-2: similarity 0.71 (online), distance 0.29
leave-1 vs password-1: similarity 0.08
```

Hints

- With unit vectors, similarity is vecs @ vecs.T.

Problem 2: Flat, HNSW and IVF

Level: Medium | **Concepts:** FAISS, recall, speed

On 20,000 random vectors, compare IndexFlatIP, IndexHNSWFlat and IndexIVFFlat: average query time and recall at 10. Try two values of efSearch and nprobe.

Sample run / expected result

```
Flat           0.30 ms/query  recall 1.00
HNSW ef=64     0.17 ms/query  recall 0.70
IVF nprobe=16  0.05 ms/query  recall 0.41
(random vectors; real embeddings give higher recall)
```

Hints

- Recall = overlap of top-10 sets / 10.
- IVF needs train() before add().

Problem 3: BM25 for Codes

Level: Easy | **Concepts:** keyword search

Use BM25 to search for "VPN-417" and "14 characters". Compare the top result with vector search.

Sample run / expected result

```
BM25 top: c9 (VPN-417 ...)
Vector top (online): c9 or c10
```

Hints

- Tokenise to lowercase words, keep codes like vpn-417 together.

Problem 4: Hybrid with RRF on Ten Questions

Level: Medium | **Concepts:** reciprocal rank fusion, hit@k, MRR

Fuse BM25 and vector rankings with RRF for ten test questions and compute hit@3 and MRR for BM25, vector and hybrid.

Sample run / expected result

```
method  hit@3  MRR      (example output; your numbers will differ)
bm25    0.80   0.71
vector  0.70   0.62
hybrid  0.90   0.78
```

Hints

- Write the expected chunk id for each question first.

Problem 5: Re-rank the Shortlist

Level: Challenging | **Concepts:** re-ranking

Retrieve the top 8 with hybrid search, re-rank with an LLM relevance score (online) or a word-overlap score (offline), and keep the top 3. Report hit@1 before and after re-ranking.

Sample run / expected result

```
hit@1 hybrid: 0.60 -> hybrid + re-rank: 0.80
```

Hints

- Ask the model for a single integer 0 to 10 per candidate.

Problem 6: Choose the Approach NEW (v4)

Level: Medium | **Concepts:** RAG, fine-tuning, long context, caching

For five scenarios (a 20-page policy asked 50 times a day; a 2,000-page manual updated weekly; a brand tone for replies; a 90,000 token contract reviewed clause by clause 5,000 times a day; a product catalogue that changes hourly), choose RAG, fine-tuning, long context, caching or a mix, and estimate monthly token cost with the placeholder prices in the notebook.

Sample run / expected result

```
20-page policy, 50/day      -> long context + cache (fits, few queries)
2,000-page manual, weekly   -> RAG (too big, changes)
brand tone                  -> fine-tuning or examples
```

Hints

- Facts that change: RAG. Style: fine-tuning or examples.
- Label every price as a placeholder.

Problem 7: Grounded Answers with a Threshold

Level: Stretch | **Concepts:** grounded generation, refusal

If the best re-ranked score is below a threshold, answer "I don't know" without calling the model. Tune the threshold on answerable and unanswerable questions and report both kinds of error.

Sample run / expected result

```
Threshold 0.25: 9/10 answerable answered, 2/2 unanswerable refused
```

Hints

- Show false refusals and false answers separately.